Lecture 5. Nonparametric Estimation of Basic Quantities for Right-Censored Data (Part 1)

Kaplan–Meier and Nelson–Aalen Estimators

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1 Setting

X is a continuous failure time with survival function $S(t)$ and cumulative hazard $H(t)$. X is subject to **non-informative right censoring**, so we observe

$$\{(T_i, \delta_i),\ i = 1, \ldots, n\}, \qquad T_i = \min(X_i, C_i), \quad \delta_i = I(X_i \le C_i).$$

Goal: estimate $S(t)$ and $H(t)$ without assuming a parametric family. We build up the estimator in four steps: discrete/continuous $\times$ without/with censoring.

2 Building Up the Estimator

2.1 Discrete X without censoring

Let X have M support points $0 < \tilde{x}_1 < \cdots < \tilde{x}_M < \infty$. With $\tilde{x}_0 = 0$, $S(\tilde{x}_0) = 1$, the product-limit representation (Lecture 2) is

$$\begin{aligned} S(x) = P(X > x) &= \prod_{\tilde{x}_j \le x} \frac{S(\tilde{x}_j)}{S(\tilde{x}_{j-1})} = \prod_{\tilde{x}_j \le x} P(X > \tilde{x}_j \mid X > \tilde{x}_{j-1}) \\ &= \prod_{\tilde{x}_j \le x} P(X > \tilde{x}_j \mid X \ge \tilde{x}_j) = \prod_{\tilde{x}_j \le x} [1 - h(\tilde{x}_j)], \qquad h(\tilde{x}_j) = P(X = \tilde{x}_j \mid X \ge \tilde{x}_j). \end{aligned}$$

(The third equality holds because there is no mass strictly between $\tilde{x}_{j-1}$ and $\tilde{x}_j$.) With i.i.d. $X_1, \ldots, X_n$, estimate

$$\hat{h}(\tilde{x}_j) = \frac{\#\{X_i = \tilde{x}_j\}}{\#\{X_i \geq \tilde{x}_j\}} = \frac{\#\text{ deaths at } \tilde{x}_j}{\#\text{ deaths at } \tilde{x}_j, \tilde{x}_{j+1}, \ldots, \tilde{x}_M}, \qquad \hat{S}(x) = \prod_{\tilde{x}_j \leq x} [1 - \hat{h}(\tilde{x}_j)].$$

$\hat{h}$ and $\hat{S}$ are the MLEs of h and S.

2.2 Discrete X with right censoring

With $T_i = \min(X_i, C_i)$, $\delta_i = I(T_i = X_i)$ and $X_i \perp\!\!\!\perp C_i$:

$$\begin{aligned} P(T_i = \tilde{x}_j, \delta_i = 1 \mid T_i \geq \tilde{x}_j) &= \frac{P(T_i = \tilde{x}_j, \delta_i = 1)}{P(T_i \geq \tilde{x}_j)} = \frac{P(X_i = \tilde{x}_j,\ C_i \geq \tilde{x}_j)}{P(X_i \geq \tilde{x}_j,\ C_i \geq \tilde{x}_j)} \\ &= \frac{P(X_i = \tilde{x}_j)P(C_i \geq \tilde{x}_j)}{P(X_i \geq \tilde{x}_j)P(C_i \geq \tilde{x}_j)} = \frac{P(X_i = \tilde{x}_j)}{P(X_i \geq \tilde{x}_j)} = h(\tilde{x}_j). \end{aligned}$$

So the hazard is identifiable from the *observed* data, which suggests

$$\hat{h}(\tilde{x}_j) = \frac{\#\{T_i = \tilde{x}_j, \delta_i = 1\}}{\#\{T_i \geq \tilde{x}_j\}}, \qquad \hat{S}(x) = \prod_{\tilde{x}_j \leq x} \Big[1 - \frac{\#\{T_i = \tilde{x}_j, \delta_i = 1\}}{\#\{T_i \geq \tilde{x}_j\}}\Big].$$

Derivation: $\hat{h}$ is the MLE under censoring

Under independent censoring, the likelihood in the hazard parameters factorizes over support points. With d_j deaths among Y_j at risk at $\tilde{x}_j$, each subject at risk either dies at $\tilde{x}_j$ (prob. h_j) or survives past it (prob. $1 - h_j$); censored subjects contribute survival up to their censoring time. Thus

$$L(h_1, \ldots, h_M) \propto \prod_j h_j^{d_j}(1 - h_j)^{Y_j - d_j}, \qquad \frac{\partial \log L}{\partial h_j} = \frac{d_j}{h_j} - \frac{Y_j - d_j}{1 - h_j} = 0 \ \Rightarrow\ \hat{h}_j = \frac{d_j}{Y_j}.$$

By invariance, $\hat{S}(x) = \prod_{\tilde{x}_j \leq x}(1 - \hat{h}_j)$ is the MLE of S.

2.3 Continuous X

Without censoring, the survival function is smooth and can be estimated by the complement of the empirical distribution function:

$$\hat{S}(t) = 1 - \mathbb{F}_n(t) = 1 - \frac{1}{n}\sum_{i=1}^{n} I(X_i \leq t).$$

With right censoring we must modify this. Two questions guide the construction: *Should the estimator jump at censoring times?* (No — no event is observed there.) *What is the contribution of a censored observation?* (It counts in the risk set Y_j for every death time up to its censoring time, then leaves.)

3 The Kaplan–Meier (Product-Limit) Estimator

Definition 5.1: Notation

- $t_1 < t_2 < \cdots < t_D$: the D distinct death times;
- $d_j = \sum_{i=1}^{n} I(T_i = t_j, \delta_i = 1)$: number of deaths at t_j;
- $Y_j = \sum_{i=1}^{n} I(T_i \geq t_j)$: number at risk (alive and under observation) just before t_j.

Theorem 5.2: Kaplan–Meier estimator

$$\hat{S}(t) = \begin{cases} 1, & t < t_1, \\ \displaystyle\prod_{t_j \leq t} \left(1 - \frac{d_j}{Y_j}\right), & t_1 \leq t. \end{cases}$$

Remark 5.3: Ties between deaths and censorings

If a death and a censoring occur at the same recorded time, the convention is that the censoring happens just *after* the death, so the censored subject is still counted in Y_j (Y_j uses $T_i \geq t_j$).

Theorem 5.4: Greenwood's variance estimator

$$\hat{V}[\hat{S}(t)] = \hat{S}(t)^2 \sum_{t_j \leq t} \frac{d_j}{Y_j(Y_j - d_j)}.$$

Derivation: delta-method derivation (Cox & Oakes, 1984)

Let $\hat{h}_j = d_j/Y_j$. Treating $d_j \mid Y_j \sim \mathrm{Bin}(Y_j, h_j)$ and the $\hat{h}_j$ as approximately uncorrelated across j,

$$\mathrm{Var}(\hat{h}_j) \approx \frac{\hat{h}_j(1-\hat{h}_j)}{Y_j}.$$

Since $\ln \hat{S}(t) = \sum_{t_j \leq t} \ln(1 - \hat{h}_j)$ and, by the delta method with $g(u) = \ln(1-u)$, $g'(u) = -1/(1-u)$:

$$\begin{aligned} \mathrm{Var}\{\ln \hat{S}(t)\} &\approx \sum_{t_j \leq t} \mathrm{Var}\{\ln(1-\hat{h}_j)\} \approx \sum_{t_j \leq t} \left(\frac{1}{1-\hat{h}_j}\right)^2 \mathrm{Var}(\hat{h}_j) \\ &\approx \sum_{t_j \leq t} \left(\frac{1}{1-\hat{h}_j}\right)^2 \frac{\hat{h}_j(1-\hat{h}_j)}{Y_j} = \sum_{t_j \leq t} \frac{\hat{h}_j}{Y_j(1-\hat{h}_j)} = \sum_{t_j \leq t} \frac{d_j/Y_j}{Y_j(Y_j-d_j)/Y_j} \\ &= \sum_{t_j \leq t} \frac{d_j}{Y_j(Y_j - d_j)}. \end{aligned}$$

A second delta-method step with $\hat{S} = \exp(\ln \hat{S})$ gives $\hat{V}[\hat{S}(t)] = \hat{S}(t)^2 \,\mathrm{Var}\{\ln \hat{S}(t)\}$.

4 The Nelson–Aalen Estimator

Theorem 5.5: Nelson–Aalen estimator and Aalen's variance

$$\tilde{H}(t) = \begin{cases} 0, & t < t_1, \\ \displaystyle\sum_{t_j \le t} \frac{d_j}{Y_j}, & t_1 \le t, \end{cases} \qquad \sigma^2_{\tilde{H}}(t) = \sum_{t_j \le t} \frac{d_j}{Y_j^2}.$$

Remark 5.6: Intuition

$H(t) = \int_0^t h(s)\,\mathrm{d}s$ accumulates the hazard; at each death time the hazard increment is estimated by $\hat{h}_j = d_j/Y_j$. Aalen's variance treats d_j as approximately Poisson with mean $Y_j h_j$, so $\mathrm{Var}(d_j/Y_j) \approx Y_j h_j / Y_j^2 \approx d_j/Y_j^2$.

4.1 Alternative estimators

- An alternative estimator of H: $\hat{H}(t) = -\log \hat{S}(t)$.
- An alternative estimator of S: $\tilde{S}(t) = \exp\{-\tilde{H}(t)\}$ (the Breslow / Fleming–Harrington estimator).
- Since $1 - x \le e^{-x}$, each KM factor $1 - d_j/Y_j \le e^{-d_j/Y_j}$, so $\hat{S}(t) \le \tilde{S}(t)$ and equivalently $\tilde{H}(t) \le \hat{H}(t)$.
- $\hat{S}(t)$ is called the "product-limit" estimator: as $n \uparrow \infty$, $D \uparrow \infty$ and the number of terms in the product $\to \infty$ for continuous X. For finite n, $D < \infty$; in the limit the product runs over the whole support of X (a *product integral*).

5 Worked Example (Small Sample)

Example 5.7: $n = 5$

Continuous X subject to right censoring, $T = \min(X, C)$, $\delta = I(X \le C)$:

T_i	0.5	1	0.75	0.25	0.75
δ_i	1	1	1	0	0

Ordered: 0.25^+, 0.5, 0.75, 0.75^+, 1. At $t = 0.75$ the censored subject is still at risk (tie convention).

t_j	d_j	Y_j	$1 - d_j/Y_j$	$\hat{S}(t_j)$	d_j/Y_j	$\tilde{H}(t_j)$	$\tilde{S}(t_j) = e^{-\tilde{H}}$
0.5	1	4	3/4	0.750	0.250	0.250	0.779
0.75	1	3	2/3	0.500	0.333	0.583	0.558
1	1	1	0	0.000	1.000	1.583	0.205

Greenwood: $\hat{V}[\hat{S}(0.5)] = 0.75^2 \cdot \frac{1}{4\cdot 3} = 0.0469$, $\hat{V}[\hat{S}(0.75)] = 0.5^2(\frac{1}{12} + \frac{1}{3\cdot 2}) = 0.0625$; at $t = 1$, $Y_j - d_j = 0$ and Greenwood is undefined (the KM drops to 0).

6 Remarks

- $\hat{S}$, $\tilde{H}$, $\hat{H}$ are right-continuous step functions even though S and H may be continuous.
- $\hat{S}$ and $\tilde{S}$ are consistent for $S(t)$ for all $t \le \tau$ such that $P(T \ge \tau) > 0$, and are asymptotically equivalent.
- $\hat{S}$ and $\tilde{H}$ are only defined for $t \le \max(T_i) = t_{\max}$. If the largest observation is censored, $\hat{S}$ does not reach 0. Some extensions for $t > t_{\max}$:
  - Efron (1967): $\hat{S}(t) = 0$;
  - Gill (1980): $\hat{S}(t) = \hat{S}(t_{\max})$;
  - Brown, Hollander & Kowar (1974): $\hat{S}(t) = \exp\{t \log[\hat{S}(t_{\max})]/t_{\max}\}$ (an exponential tail).

7 The 6-MP Leukaemia Trial

Data. 42 children with acute leukaemia in complete or partial remission were matched in pairs by remission status and randomized within the pair to 6-mercaptopurine (6-MP) or placebo maintenance therapy, then followed until relapse or end of study. Times to relapse (weeks) for the 21 **6-MP** patients ($^+$ = censored):

$$6,\ 6,\ 6,\ 6^+,\ 7,\ 9^+,\ 10,\ 10^+,\ 11^+,\ 13,\ 16,\ 17^+,\ 19^+,\ 20^+,\ 22,\ 23,\ 25^+,\ 32^+,\ 32^+,\ 34^+,\ 35^+.$$

Placebo times (all observed): $1, 1, 2, 2, 3, 4, 4, 5, 5, 8, 8, 8, 8, 11, 11, 12, 12, 15, 17, 22, 23$.

Example 5.8: Kaplan–Meier estimate for the 6-MP group

t_j	d_j	Y_j	$\hat{S}(t) = \prod(1 - d_j/Y_j)$	$\sum \frac{d_j}{Y_j(Y_j - d_j)}$	$\hat{S}(t)^2 \sum \cdots$
6	3	21	$[1 - \frac{3}{21}] = 0.857$	$\frac{3}{21\times 18} = 0.0079$	0.0058
7	1	17	$0.857(1 - \frac{1}{17}) = 0.807$	$0.0079 + \frac{1}{17\times 16} = 0.0116$	0.0076
10	1	15	$0.807(1 - \frac{1}{15}) = 0.753$	$0.0116 + \frac{1}{15\times 14} = 0.0164$	0.0093
13	1	12	$0.753(1 - \frac{1}{12}) = 0.690$	$0.0164 + \frac{1}{12\times 11} = 0.0240$	0.0114
16	1	11	$0.690(1 - \frac{1}{11}) = 0.628$	$0.0240 + \frac{1}{11\times 10} = 0.0330$	0.0130
22	1	7	$0.628(1 - \frac{1}{7}) = 0.538$	$0.0330 + \frac{1}{7\times 6} = 0.0569$	0.0164
23	1	6	$0.538(1 - \frac{1}{6}) = 0.448$	$0.0569 + \frac{1}{6\times 5} = 0.0902$	0.0181

Note how censorings reduce Y_j without producing a jump: e.g. between $t = 7$ ($Y = 17$) and $t = 10$ ($Y = 15$) the subject censored at 9^+ leaves the risk set.

Example 5.9: Nelson–Aalen estimate for the 6-MP group

t	$\tilde{H}(t) = \sum d_j/Y_j$	$\sigma^2_H = \sum d_j/Y_j^2$	SE
$0 \le t < 6$	0	0	0
$6 \le t < 7$	$3/21 = 0.1428$	$3/21^2 = 0.0068$	0.0825
$7 \le t < 10$	$0.1428 + 1/17 = 0.2017$	$0.0068 + 1/17^2 = 0.0103$	0.1015
$10 \le t < 13$	$0.2017 + 1/15 = 0.2683$	$0.0103 + 1/15^2 = 0.0147$	0.1212
$13 \le t < 16$	$0.2683 + 1/12 = 0.3517$	$0.0147 + 1/12^2 = 0.0217$	0.1473
$16 \le t < 22$	$0.3517 + 1/11 = 0.4426$	$0.0217 + 1/11^2 = 0.0299$	0.1729
$22 \le t < 23$	$0.4426 + 1/7 = 0.5854$	$0.0299 + 1/7^2 = 0.0503$	0.2243
$23 \le t < 35$	$0.5854 + 1/6 = 0.7521$	$0.0503 + 1/6^2 = 0.0781$	0.2795

Comparison. Plotting $\tilde{S} = e^{-\tilde{H}}$ against $\hat{S}$ (K&M Fig. 4.1A), the Nelson–Aalen-based curve lies slightly *above* the KM curve at every step, and correspondingly $\hat{H} = -\log \hat{S}$ lies slightly above $\tilde{H}$; the gap widens in the tail where Y_j is small (e.g. at $t = 23$: $\hat{S} = 0.448$ vs. $\tilde{S} = e^{-0.752} = 0.471$).